

Experiment No-1

Student Name: HARSH

Branch: MCA

Semester: 3rd

Subject Name: Machine Learning Lab

UID: 24MCA20045

Section/Group: 24MCA-1(B)

Date of Performance: 25/07/2025

Subject Code: 24CAP-702

Aim:

1. Implementation of Python Basic Libraries such as
 - Math
 - NumPy
 - Matplotlib
 - Seaborn
 - SciPy.
2. Creation and loading different Datasets in Python using Jupyter Notebook.
3. Write a program to implement Linear regression on Jupyter notebook.
 - a. Download different datasets from Kaggle or UCI ML Repository.
 - b. Import all the necessary modules for the datasets.
 - c. Find out the best fit line along with the MSE, RMSE.

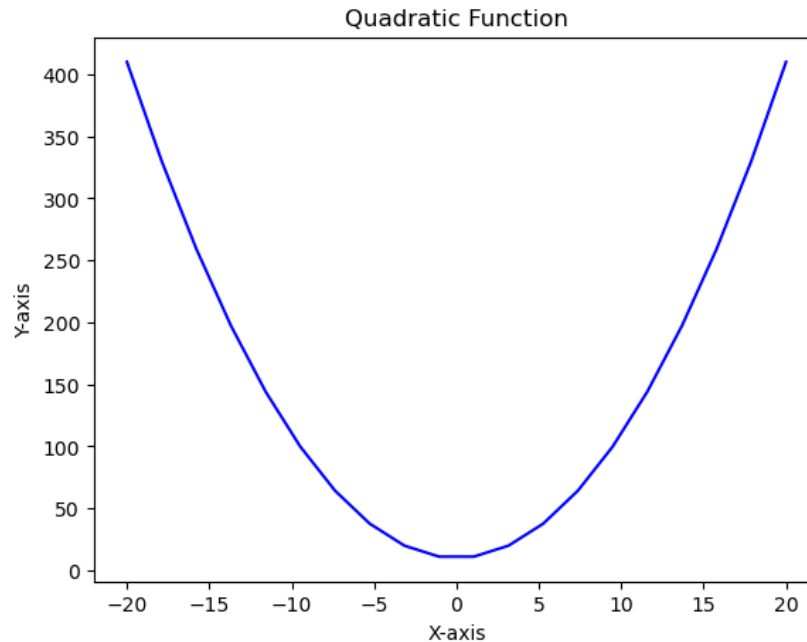
Code:

```
1.
import math
p=int(input("Enter a number: "))
q=math.sqrt(p)
print(q)
import matplotlib.pyplot as plt
import numpy as np
x=np.linspace(-20,20,20)
y=x**2+10
fig1=plt.figure()
axes= fig1.add_axes([0.2,0.2,0.8,0.8])
axes.plot(x,y,'b')
axes.set_xlabel('X-axis')
```

```
axes.set_ylabel('Y-axis')
axes.set_title('Quadratic Function')
```

Output:

```
Enter a number: 9
3.0
[3]: Text(0.5, 1.0, 'Quadratic Function')
```



2.

```
import pandas as pd
df=pd.read_csv("C:\Users\harsh\Downloads\archive\ movies_metadata.csv")
df
```

Output:

```
[2]:
```

	gender	race/ethnicity	parental level of education	lunch	test preparation course	math score	reading score	writing score
0	female	group B	bachelor's degree	standard	none	72	72	74
1	female	group C	some college	standard	completed	69	90	88
2	female	group B	master's degree	standard	none	90	95	93
3	male	group A	associate's degree	free/reduced	none	47	57	44
4	male	group C	some college	standard	none	76	78	75
...	...	...	...	...	...	...	...	...
995	female	group E	master's degree	standard	completed	88	99	95
996	male	group C	high school	free/reduced	none	62	55	55
997	female	group C	high school	free/reduced	completed	59	71	65
998	female	group D	some college	standard	completed	68	78	77
999	female	group D	some college	free/reduced	none	77	86	86

1000 rows × 8 columns

3.

Code:

```
import numpy as np
import matplotlib.pyplot as plt

X = np.array([1, 2, 3, 4, 5])
Y = np.array([2, 4, 5, 4, 5])

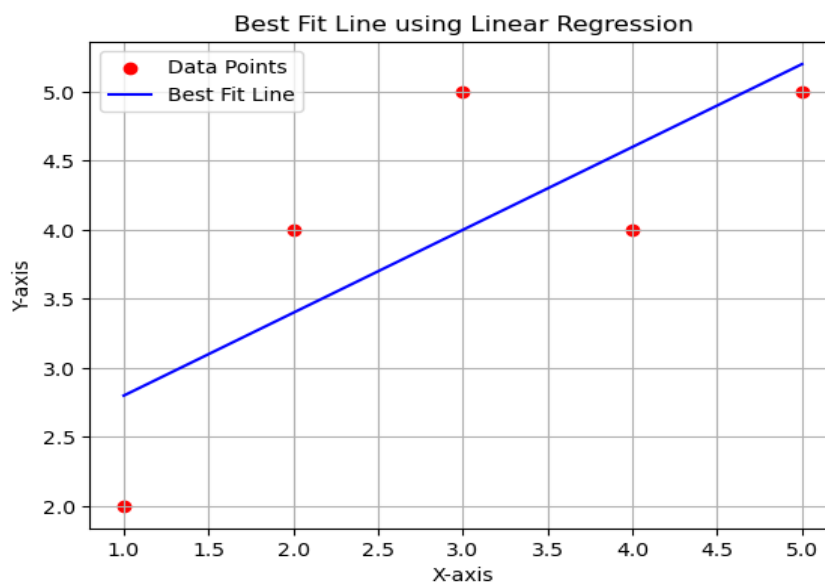
mean_X=np.mean(X)
mean_Y=np.mean(Y)

numerator = np.sum((X - mean_X) * (Y - mean_Y))
denominator = np.sum((X - mean_X)**2)
m = numerator / denominator

c = mean_Y - m * mean_X
Y_pred = m*X + c

plt.scatter(X, Y, color='red', label='Data Points')
plt.plot(X, Y_pred, color='blue', label='Best Fit Line')
plt.xlabel("X-axis")
plt.ylabel("Y-axis")
plt.title("Best Fit Line using Linear Regression")
plt.legend()
plt.grid(True)
plt.show()
```

Output:



Learning Outcomes:

1. Understood and utilize basic Python libraries such as math, numpy, matplotlib.
2. Learned how to create, import, and load datasets from external source like Kaggle.
3. Implement Linear Regression from scratch and using libraries, understanding the step-by-step mathematical logic and code structure.
4. Analyzed data trends using a best fit line by calculating performance metrics like MSE (Mean Squared Error) and RMSE (Root Mean Squared Error) to evaluate model accuracy.

Evaluation Grid:

Sr. No.	Parameters	Marks Obtained	Maximum Marks
1.	Demonstration and Performance		12
2.	Worksheet		8
3.	Via		10